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27th Jan 2026 / DAG360

⇒ Informal disc. on solution methods

-
$$\begin{aligned} \min \quad & f(x) \leftarrow \text{obj. func.} \\ \text{s.t.} \quad & x \in \mathcal{X} \leftarrow \text{constant} \\ & \hookrightarrow \text{optimization variable} \end{aligned}$$

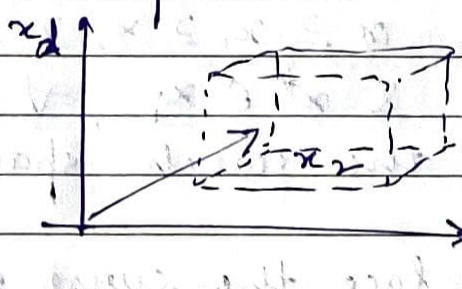
⇒ check for all x 's, find the min.
↳ but this process is not feasible always
like $x \in \mathbb{R}^d$ & contains ∞ pts.

⇒ give up on ∞ pts. → come to finite pts.

⇒ Boundary that has minima. $\min_x x^2$
(check the boundary) $|x| \leq 5$

⇒ check stationary pts. $\min x^2$
 $x \geq 5$

⇒ grid the space



G^d
exponential
nature
curse of
dimensionality

eg. 10 warehouse + 1 ⇒ int increases
by G factors.

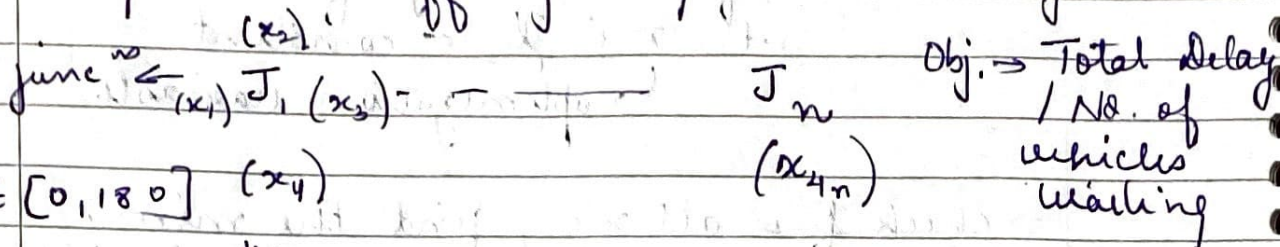
- so for what we have discussed our
0th order methods.

$x_1, \dots, x_n$ → simultaneous perturbations
 $f(x_1), \dots, f(x_n)$ (stochastic approximation)

⇒ examine the critical pts.

assume $\rightarrow f$ is m a functional form.

- eg. in a traffic juncⁿ / network congestion.



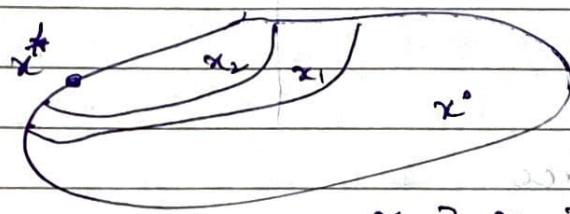
$$x_i \in [0, 180] \quad (x_4)$$

$$x \in \mathbb{R}^{4n}$$

we don't know what the f is, we can only calculate what f is. It's analytical form is not known. Therefore, they accept 0th order methods.

⇒ Shrink / eliminaⁿ method

(ellipsoidal method)



$$x \geq x_1 \geq x_2 \geq \dots$$

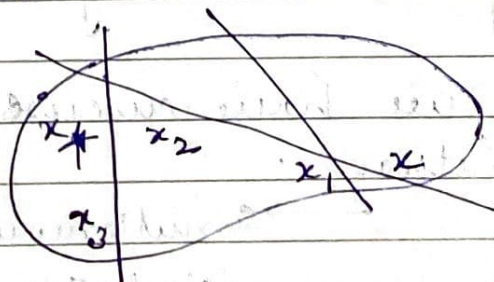
$$\& x^* \in x_i, \forall i$$

we shrink the search space progressively.

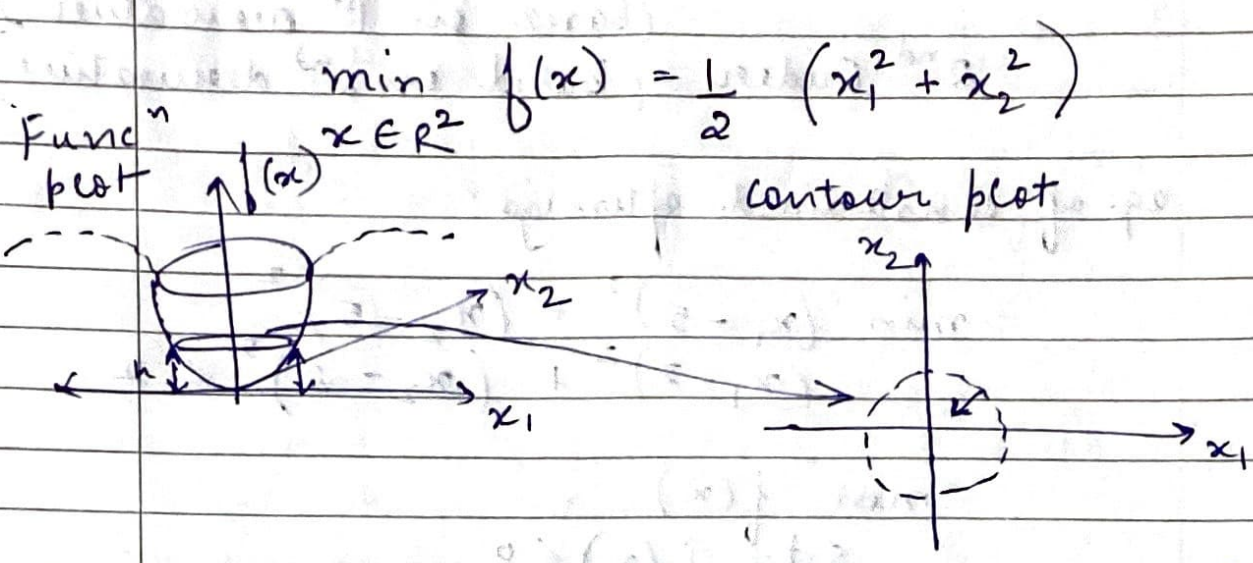
we again face the curse of dimensionality

⇒ slice the search space

same issue
(cutting plane method)



→ Direction-based methods.



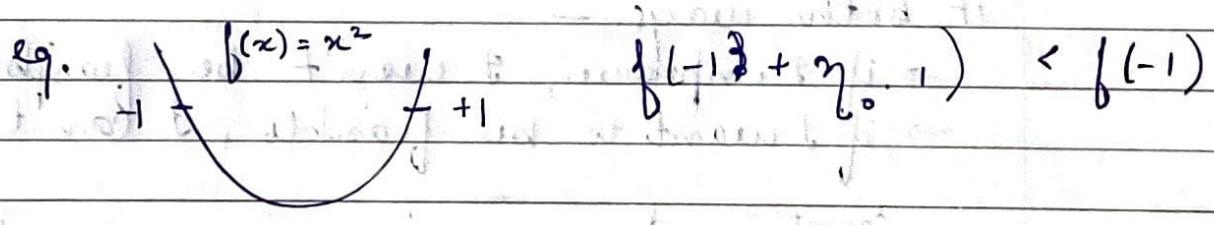
improve the direcⁿ → pick any pt. $x_i \in \mathbb{R}^d$, then the next pt. is in improving direcⁿ

$f(x_i + \eta u_{\text{imp}}) < f(x_i)$ at $x \in \mathbb{R}^d$ weak $\eta = 0.1, 0.2, \dots$

- A direcⁿ is improving, if there exists a $\eta_0 > 0$ s.t. for every $\eta \in (0, \eta_0)$.

scaler

$$f(x_i + \eta u_{\text{imp}}) < f(x_i)$$



- many a times, we can guess the value of good candidates (η). Try $\eta_i \in (0, \eta_0)$

→ Direcⁿ 1st Order Methods

2nd Order (based on 1st derivative)
(based on 2nd derivative)

eq. of constrained optimizaⁿ

$$\min (x_1 - 5)^2 + (x_2 - 5)^2$$

$$(x_1 - 2)^2 + (x_2 - 2)^2 \leq 2$$

$$\min f(x)$$

$$\text{s.t. } g(x) \leq 0$$

$$g(x) = (x_1 - 2)^2 + (x_2 - 2)^2 - 2$$

feasible direcⁿ → associated with $g(x)$

at $x \in \mathbb{R}^d$, $u \in \mathbb{R}^d$ is a feasible direcⁿ if there exists a γ_0 s.t. for every $\gamma \in (0, \gamma_0)$

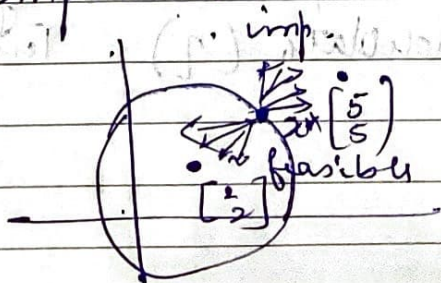
$$g(x + \gamma u_{\text{feas}}) \leq 0 \quad (g(x) < 0)$$

At the constraint optimum, we can't have it both ways -

→ if I improve, I won't be feasible

→ if I want to be feasible, I can't improve

$x^{\text{const}} \in \mathbb{R}^d$ no direcⁿ is both feasible or improves



- constrained problem

↓
convert constraints into penalty

$$L(x, \mu, \lambda) = f(x) + \sum_{i=1}^n \lambda_i g_i(x) +$$

$$\sum_{j=1}^m H_j h_j(x)$$

$\lambda_i \geq 0$ $H_j \in \mathbb{R}$

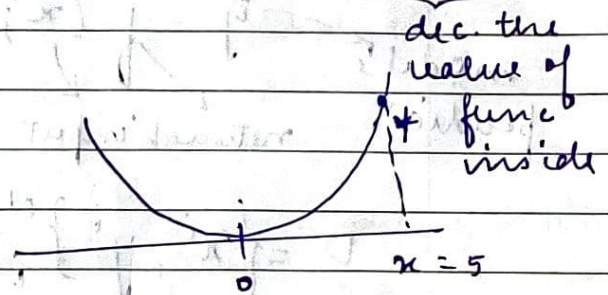
$$g(\lambda, \mu) = \min_x L(x, \lambda, \mu)$$

Primal-dual methods

eg. $\min_{x \geq 5} x^2 \rightarrow L(x, \lambda) = x^2 + \lambda(5-x)$

Lagrangian

$$\equiv \min_{x \geq 5} x^2$$



$$L(x, 0) = x^2$$